25900363 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT205 - MICROCONTROLLER PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What is the input voltage for the power supply pins on a Raspberry Pi? What are the types of power output pins available on a Raspberry Pi? [K] / [CO1]

2. What is the function of print() in Python programming? [U] / [CO1]

3. Can you list the main elements that make up the structure of a Python program? [K] / [CO1]

4. What will be the output of the following codes? [U] / [CO1]

   1.
   ```
   x = 3
   y = 2
   print(x ** y)
   ```

   2.
   ```
   a = 23
   b = 4
   print(a // b)
   ```

5. What are keywords in Python, and provide examples of keywords? [U] / [CO1]

6. Briefly explain the requirements for setting up Raspberry Pi. [K] / [CO1]

7. What does a tuple represent as a sequence in Python? [K] / [CO1]

8. Explain an example use of digital write in controlling an LED. [K] / [CO2]

9. What is the purpose of the int() function in Python? Explain with an example. [U] / [CO2]

10. Given the string = "Binary:101010", write a Phyton program to extract only the binary number from it. [U] / [CO2]

11. Which Python library is used to play simple audio files? [K] / [CO2]

12. What are the fundamental steps in creating a Tkinter GUI? [K] / [CO3]

13. Define pack in Tkinter geometry managers. [K] / [CO3]

14. What is the purpose of the label.place(x=50, y=50) line in the given code? [K] / [CO3]

[2x10 = 20]

Part B

Short Essay questions

Answer any **6** questions

Each question carries **5** marks

15. What is Raspberry Pi, and what are its key features and applications? Name some popular models of Raspberry pi. [U] / [CO1]

16. Explain the difference between integers, floats, and complex numbers in Python. Provide examples of each. [K] / [CO1]

17. Explain how pin numbering affects hardware-software synchronization. [U] / [CO2]

18. Discuss the advantages and limitations of GPIO in interfacing external devices. [U] / [CO2]

19. Discuss the role of GPIO libraries in modular and reusable code development. [U] / [CO2]

20. Give an overview of Tkinter 's funtionality . [U] / [CO3]

21. How can you customize the properties of Tkinter widgets? [K] / [CO3]

22. Can you explain the functionality of Tkinter widgets, with practical examples? [K] / [CO3]

[5x6 = 30]